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```
name =rocky kumar gautam
email id =rockygaugtam09890@gmail.com
assigment name= python basics
```

1 Explain the key features of Python that make it a popular choice for programming

Easy to use and learn. Python has a simple syntax that is easy to learn. ... Open Source. Python is an open-source language. ... Large Standard Library. ... Dynamically Typed. ... Large Community Support. ... Portable. ... Platform Independent. ... Scalable

2 Describe the role of predefined keywords in Python and provide examples of how they are used in a program

Keywords in Python are predefined words with special meanings to the interpreter. Each of them is meant to perform a specific task. On the other hand, identifiers are user-defined words given to different parts of Python programming. We can assign identifier names to a variable, function, or class. Keywords in Python are case-sensitive.

All the keywords must be in lowercase except True and False.

They may vary with Python versions.

Keywords have specific meanings and predefined uses, which can't be altered.

They are a part of Python syntax and can't be changed.

They must be according to the syntax rules in Python.

Keywords can't be used as identifiers.

Some words can be used as keywords in specific contexts and as identifiers in other cases. Keyword

Description

True

Represents the boolean value True.

False

Represents the boolean value false.

None

A special constant that represents a null value. 0 or empty list doesn't compute None.

and

A logical operator that combines conditional statements and returns True if both the conditions are true; else returns False.

or

A logical operator that combines conditional statements and returns True if one condition is true.

not

A logical operator that inverts the truth value so it returns True if the condition is false.

is

Checks if two values are identical.

in

Checks if a value is present in a range, list, tuple, etc.

if

Makes a conditional statement.

else

A conditional statement used with if and elif conditional statements and executed if the given condition is not true.

elif

A conditional statement used with an if statement and executed when the previous condition is not true.

del

Deletes a reference to an object.

for

Creates a for loop.

while

Creates a while loop.

break

Terminates the current closest enclosing loop.

continue

Continues the next iteration of the current loop.

import

Imports a module.

from

Imports specific parts of a module.

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3 Compare and contrast mutable and immutable objects in Python with example

Mutable Data Type – A mutable data type is one whose values can be changed. Example: List, Dictionaries, and Set

Immutable Data Type – An immutable data type is one in which the values can't be changed or altered. Example: String and Tuples

4 Discuss the different types of operators in Python and provide examples of how they are used

Python language supports various types of operators, which are: Arithmetic Operators Comparison (Relational) Operators Assignment Operators Logical Operators Bitwise Operators Membership Operators Identity Operators

a = 21 b = 10

✓ Addition

print ("a + b : ", a + b)

Subtraction

print ("a - b : ", a - b)

Multiplication

print ("a * b : ", a * b)

Division

print ("a / b : ", a / b)

Modulus

print ("a % b : ", a % b)

Exponent

print ("a ** b : ", a ** b)

Floor Division

print ("a // b : ", a // b)

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5 Explain the concept of type casting in Python with examples

The conversion of one data type into the other data type is known as type casting in python or type conversion in python. Python supports a wide variety of functions or methods like: `int()`, `float()`, `str()`, `ord()`, `hex()`, `oct()`, `tuple()`, `set()`, `list()`, `dict()`, etc. for the type casting in python.

6 How do conditional statements work in Python? Illustrate with examples

If the condition is True, the code block indented below the if statement will be executed. If the condition is False, the code block will be skipped. Here's an example of how to use an if statement to check if a number is positive: `num = 5` if `num > 0`: `print("The number is positive.")`

7 Describe the different types of loops in Python and their use cases with examples

While loop

Repeats the given statement or group of statements until the given condition is true. The while loop programs in Python test the condition before executing the loop body. `count = 1` while `count <= 5`: `print(count)` `count += 1`

For Loops in Python

We use the for loop program in Python for sequential traversal. It is designed to iterate over a sequence, such as a list, dictionary, tuple, set, and string. Iteration is the process of traversing a sequence.

It works more like an iterator method in object-oriented programming than the for loop in other programming languages. The for-in loop in Python is similar to the for-each loop in other programming languages

Example list of numbers

```
numbers = [1, 2, 3, 4, 5]
```

Using a for loop to iterate through the list

```
for number in numbers: print(number)
```